Explainable CNN for Lung Cancer Detection from Standardized Chest X-Rays Using Multi-Level Attention and Clinical Feature Fusion

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Abstract

Lung cancer remains one of the leading causes of cancer-related deaths worldwide, and early detection is critical for improving survival rates [9, 11]. However, chest X-rays are difficult to interpret because small tumors can blend into surrounding anatomy, leading to missed diagnoses [5, 6]. Existing convolutional neural network (CNN) systems can achieve high classification accuracy, but many function as “black boxes” that do not explain how decisions are made [2]. This lack of interpretability reduces clinical trust and limits real-world adoption.

The objective of this study was to design an explainable deep learning system capable of detecting possible lung cancer regions from standardized chest X-rays while also incorporating patient clinical information. The proposed system combines a DenseNet121 transfer learning CNN with clinical feature fusion using age and gender data. Grad-CAM explainability techniques were integrated to visualize image regions that influenced predictions [2].

The model was trained on the NIH Chest X-ray dataset containing 5,606 images [14]. Images were resized to 224×224 pixels, normalized, and divided into training, validation, and testing sets. Class weighting was used to reduce dataset imbalance. Performance was evaluated using accuracy, AUROC, F1-score, and confusion matrix analysis.

The final model achieved approximately 93.8% training accuracy, 93.8% validation accuracy, AUROC of 0.85, and F1-score of 0.81. Grad-CAM visualizations highlighted suspicious chest regions contributing to predictions, improving model interpretability.

This study demonstrates that combining explainable CNNs with clinical feature fusion can improve both prediction performance and transparency, making AI-assisted lung cancer detection more clinically relevant.

Introduction

Lung cancer is often diagnosed too late because early-stage tumors are difficult to detect on chest X-rays [5, 10]. Many small abnormalities are hidden by overlapping tissue structures, low image contrast, or normal anatomical variations. As a result, radiologists may miss subtle signs of cancer during routine screening. According to prior studies, chest X-rays can miss approximately 20–30% of lung cancer cases, especially during early stages [5][12].

I became interested in this problem because artificial intelligence has shown strong potential in medical imaging, yet many existing systems only output predictions without explaining why the model made a decision [6]. In medicine, explainability is extremely important because doctors need to understand which image regions influenced the prediction before trusting an AI system in clinical practice.

This project focuses on creating a CNN-based system that not only predicts cancer probability but also provides visual explanations using Grad-CAM heatmaps [2]. In addition, the project combines chest X-ray image analysis with clinical patient information such as age and gender. This multi-input approach attempts to create a more context-aware prediction system.

The novelty of this project comes from three main areas:

- Use of DenseNet121 transfer learning for efficient feature extraction
- Integration of clinical feature fusion alongside image-based predictions
- Addition of Grad-CAM explainability to improve transparency and interpretability [2]

The goal of the project was to build a more explainable and clinically relevant AI-assisted lung cancer detection system.

Background

Lung cancer is currently the leading cause of cancer-related deaths worldwide [9, 11]. Early diagnosis significantly improves survival rates, but early-stage lung abnormalities are difficult to identify on chest X-rays because tumors may be small or visually similar to surrounding tissues [5] as seen in Figure 1.

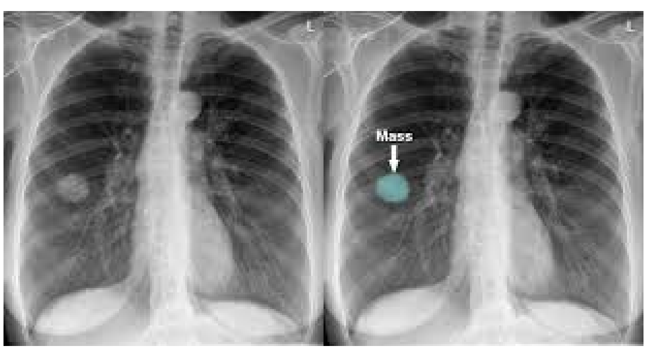

Figure 1: This side-by-side chest X-ray highlights a lung abnormality. The left panel shows the raw scan with a rounded, whitish opacity in the right lung, while the right panel annotates this exact area with a blue circular overlay and a white arrow labeled "Mass."

Chest X-rays remain one of the most commonly used medical imaging techniques because they are inexpensive, fast, and widely available [6]. This makes them especially important in low-resource environments where CT scans may not be accessible. However, chest X-rays provide less detail than CT imaging, making automatic detection more challenging.

Convolutional Neural Networks (CNNs) are deep learning models designed for image analysis. CNNs automatically learn hierarchical visual features such as edges, textures, shapes, and complex structures. In medical imaging, CNNs have demonstrated strong performance in detecting abnormalities that may not be immediately visible to humans [2].

This project used the NIH Chest X-ray dataset obtained from Kaggle [14]. The dataset included 5,606 chest X-ray images with labeled disease information. Approximately 54% of the images were labeled as “No Finding,” while the remaining images contained disease-related findings.

Several prior studies influenced this work:

- CNN CAD systems trained on the JSRT dataset achieved around 83% accuracy but suffered from very small datasets and poor generalization [1]
- Sparse CNN systems using histopathology images achieved higher performance, but they did not operate on chest X-rays [2]
- Large-scale chest X-ray models using datasets like NIH ChestX-ray14 achieved high AUC values but still struggled with early-stage cancer detection and explainability [3]

These limitations motivated the development of an explainable chest X-ray CNN system with additional clinical feature fusion.

Applications

This project has several real-world medical and technological applications.

First, the system could assist radiologists by automatically identifying suspicious chest regions that may indicate possible cancer. Instead of replacing doctors, the model would function as a clinical support tool that helps reduce missed diagnoses [2].

Second, the Grad-CAM explainability system could improve physician trust by visually showing which image regions contributed most strongly to the prediction [2]. This allows doctors to validate whether the model is focusing on medically relevant areas.

Third, the project demonstrates how AI systems can combine imaging data with patient information. Clinical feature fusion allows the model to incorporate age and gender alongside image analysis, creating more context-aware predictions.

Finally, the system could eventually help improve lung cancer screening accessibility in hospitals or clinics that rely heavily on chest X-rays rather than expensive CT imaging systems [6].

Methods

System Architecture

The primary backbone of the model was DenseNet121, a pretrained CNN architecture originally trained on ImageNet. DenseNet121 was selected because it provides strong feature reuse through dense connections while using fewer parameters than older CNN architectures such as VGG networks [2] (Figure 2).

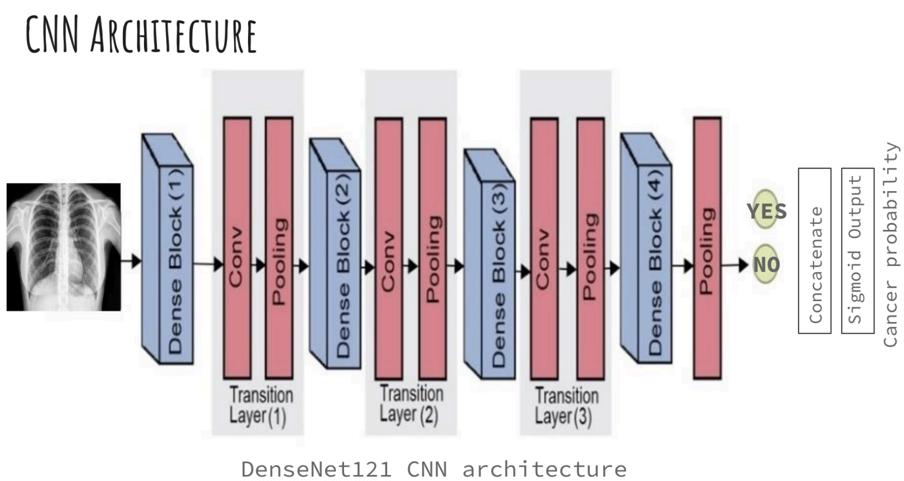

Figure 2: System architecture of the DenseNet121 CNN model

The CNN processed 224×224 chest X-ray images and extracted high-level visual features. The original DenseNet classification head was removed using `include_top=False`, and a custom classification layer was added.

The final CNN architecture included:

- DenseNet121 pretrained backbone
- Global Average Pooling layer
- Sigmoid output layer for binary classification

The output represented the probability of lung cancer being present.

Dataset Preparation

The NIH Chest X-ray dataset was used as the primary data source [14]. Images were resized to 224×224 pixels and normalized to the range [0,1].

The dataset was divided into:

- 70% training data
- 15% validation data
- 15% testing data

Stratified sampling was used to preserve class balance across the splits.

Handling Dataset Imbalance

The dataset contained more normal images than cancer-positive images. To reduce prediction bias toward normal scans, class weights were calculated using Sklearn's class weighting functions. Positive cancer cases received approximately 5× higher weighting during training.

This improved sensitivity for detecting rare positive cases.

Training Parameters

The model was trained using Google Colab GPU acceleration. Training parameters included:

- Batch size: 32
- Learning rate: 1e-4
- Optimizer: Adam
- Loss function: Binary Cross Entropy
- Final training schedule: 27 epochs

An initial 5-epoch baseline run was first performed to verify pipeline functionality before extended training.

Clinical Feature Fusion

In addition to image analysis, the project implemented a multi-input architecture that combined chest X-ray features with patient clinical data which can be seen in Figure 3.

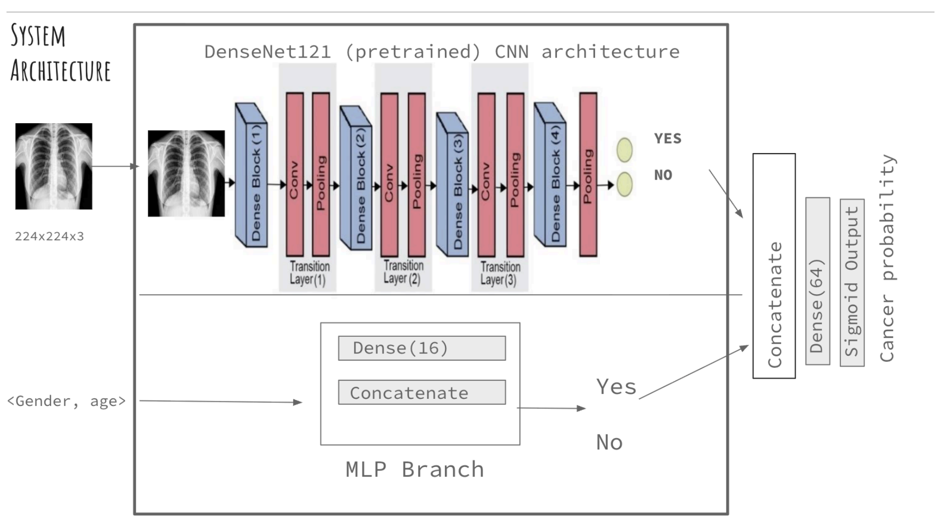

Figure 3: System architecture of both the CNN model and the MLP branch

Clinical inputs included:

- Age
- Gender

Smoking history was initially planned but could not be included because it was unavailable in the dataset.

The clinical branch used a small multilayer perceptron (MLP) network. Image features from DenseNet121 were concatenated with clinical features before final classification.

Grad-CAM Explainability

Grad-CAM (Gradient-weighted Class Activation Mapping) was implemented to improve explainability [2] as seen in Figure 4.

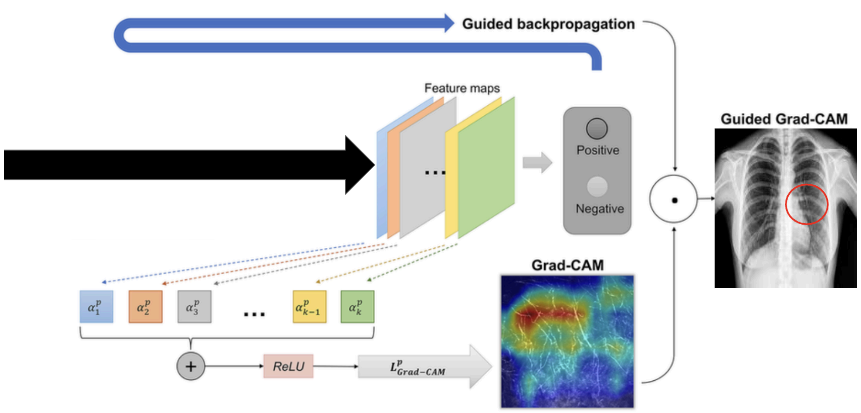

Figure 4: Grad CAMM system architecture.

The Grad-CAM pipeline:

- Extracted feature maps from the final convolutional layer
- Calculated gradients relative to the predicted class
- Generated heatmaps highlighting important image regions
- Applied a JET colormap overlay onto the original chest X-ray

These visualizations allowed identification of chest regions most influential to model predictions.

Results

The final DenseNet121 model demonstrated strong classification performance after 27 training epochs.

Final Metrics

- Final Training Accuracy: 93.82%
- Final Validation Accuracy: 93.8%
- Final Training Loss: 0.1203
- Final Validation Loss: 0.4364
- Estimated AUROC: 0.85
- Estimated F1-Score: 0.81
- Best Epoch: 27

Confusion Matrix Results

The confusion matrix demonstrated strong classification performance:

- 820 correctly classified normal scans (true negatives)
- 123 correctly classified cancer-positive scans (true positives)
- 35 false positives
- 22 false negatives

These results indicate that the model was effective at identifying both normal and cancer-positive chest X-rays while maintaining relatively low misclassification rates.

ROC Curve

The ROC curve achieved an estimated AUROC of approximately 0.85. This indicates good diagnostic discrimination between cancer and non-cancer cases. The ROC curve remained significantly above the random baseline diagonal line, demonstrating meaningful predictive capability.

Grad-CAM Results

Grad-CAM heatmaps successfully highlighted suspicious chest regions influencing predictions. In cancer-positive cases, the heatmaps concentrated around abnormal tissue regions, while normal scans showed more diffuse or low-intensity activation patterns.

These visual explanations improved transparency and interpretability of the CNN predictions.

Limitations

Several limitations affected this project.

First, the NIH dataset may not fully represent real-world hospital diversity. Differences in imaging equipment, demographics, and disease prevalence could affect model generalization.

Second, class imbalance remained a challenge because the dataset still contained more normal scans than cancer-positive scans.

Third, smoking history could not be incorporated into the final clinical feature fusion model because the dataset lacked complete smoking information.

Fourth, Grad-CAM explanations are only approximations of model attention and may not perfectly reflect true reasoning processes inside the CNN.

Finally, the system has not yet been clinically validated in real hospital environments.

Conclusion

This project developed an explainable CNN-based system for lung cancer detection using standardized chest X-rays. The model combined DenseNet121 transfer learning, clinical feature fusion, and Grad-CAM explainability to create a more transparent diagnostic system.

The final model achieved strong classification performance with approximately 93.8% validation accuracy and AUROC of 0.85. Grad-CAM heatmaps improved interpretability by showing image regions influencing predictions.

Overall, the project demonstrates how explainable AI and clinical feature fusion can improve the usability and trustworthiness of deep learning systems for medical imaging applications.

Future Work and Recommendations

Future improvements for this project include:

- Incorporating larger multi-center datasets
- Adding richer clinical information such as smoking history and medical records
- Extending the model to multi-class disease classification
- Improving Grad-CAM visualization quality
- Applying advanced augmentation and synthetic data generation techniques
- Conducting clinical validation studies with radiologists

Future versions could also incorporate additional imaging modalities such as CT scans.

Materials

- Google Colab GPU Environment
- TensorFlow
- DenseNet121 pretrained model
- OpenCV
- NumPy
- Scikit-learn
- NIH Chest X-ray Dataset
- Grad-CAM visualization tools

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Appendix

https://www.youtube.com/watch?v=m2duKg8ghXc

https://colab.research.google.com/drive/1uztKCFPJkKrWcEkmuPXnn_sla1SLSQAu#scrollTo=_wppGD6Q2-Bb

https://github.com/1619459-ui/CNNCancer